

1 Supplementary Material

Table S1. Performance (mean $\pm$ standard deviation) of selected loss functions across GWN, LSTM, TCN, and MLP architectures at the $t + 1$ prediction horizon. Loss functions are grouped into non-weighted, existing weighted, and novel weighted categories. Bold onset recall values denote the highest-performing model for each loss function, while underlined values denote the second-best performer. The ** marker indicates the overall best onset recall across all model-loss combinations.

Loss Type	GWN				LSTM				TCN				MLP			
	Onset Recall	Overall			Onset Recall	Overall			Onset Recall	Overall			Onset Recall	Overall		
		Recall	Precision	F1		Recall	Precision	F1		Recall	Precision	F1		Recall	Precision	F1
BCE	0.091 ± 0.057	0.646 ± 0.198	0.785 ± 0.102	0.683 ± 0.161	0.124 ± 0.023	0.750 ± 0.015	0.796 ± 0.019	0.772 ± 0.009	<u>0.107</u> ± 0.043	0.475 ± 0.099	0.801 ± 0.060	0.587 ± 0.093	0.020 ± 0.024	0.124 ± 0.070	0.629 ± 0.069	0.202 ± 0.089
PBCE	0.063 ± 0.054	0.636 ± 0.150	0.834 ± 0.055	0.706 ± 0.109	0.117 ± 0.025	0.740 ± 0.021	0.799 ± 0.022	0.768 ± 0.011	<u>0.113</u> ± 0.042	0.481 ± 0.085	0.790 ± 0.055	0.591 ± 0.075	0.010 ± 0.012	0.118 ± 0.039	0.596 ± 0.068	0.194 ± 0.056
WBCE	0.817 ± 0.034	0.957 ± 0.008	0.436 ± 0.042	0.597 ± 0.040	<u>0.783</u> ± 0.101	0.942 ± 0.053	0.461 ± 0.041	0.617 ± 0.037	0.640 ± 0.063	0.912 ± 0.030	0.452 ± 0.043	0.603 ± 0.039	0.521 ± 0.128	0.885 ± 0.039	0.467 ± 0.100	0.602 ± 0.081
Focal	<u>0.636</u> ± 0.057	0.915 ± 0.013	0.436 ± 0.045	0.589 ± 0.040	0.750 ± 0.032	0.941 ± 0.007	0.431 ± 0.013	0.591 ± 0.011	0.500 ± 0.076	0.816 ± 0.105	0.491 ± 0.047	0.609 ± 0.056	0.585 ± 0.122	0.890 ± 0.108	0.446 ± 0.089	0.585 ± 0.099
WPBCE	0.824 ± 0.023	0.959 ± 0.005	0.444 ± 0.029	0.606 ± 0.026	<u>0.796</u> ± 0.087	0.952 ± 0.021	0.446 ± 0.044	0.606 ± 0.040	0.722 ± 0.050	0.935 ± 0.012	0.400 ± 0.043	0.559 ± 0.041	0.714 ± 0.071	0.933 ± 0.016	0.411 ± 0.074	0.566 ± 0.075
WHinge	0.844** ± 0.013	0.962 ± 0.003	0.444 ± 0.021	0.603 ± 0.020	<u>0.785</u> ± 0.093	0.946 ± 0.046	0.462 ± 0.040	0.619 ± 0.034	0.641 ± 0.088	0.902 ± 0.102	0.468 ± 0.046	0.612 ± 0.061	0.689 ± 0.066	0.927 ± 0.015	0.448 ± 0.050	0.602 ± 0.046
WSqHinge	0.831 ± 0.022	0.960 ± 0.005	0.444 ± 0.025	0.605 ± 0.022	<u>0.823</u> ± 0.015	0.959 ± 0.004	0.446 ± 0.019	0.604 ± 0.017	0.688 ± 0.074	0.927 ± 0.018	0.446 ± 0.058	0.599 ± 0.053	0.719 ± 0.070	0.932 ± 0.027	0.448 ± 0.053	0.603 ± 0.048